Ludovica Melodia

ENGINEERING PORTFOLIO

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Motion Station: Boston Children's Museum Exhibit

January – May 2026

GE 1502, Northeastern University · Four-person team · Client: Boston Children's Museum

Our four-person team built an interactive exhibit for the Boston Children's Museum that teaches children ages 4–8 about gear ratios: an Italy-themed, *Cars*-inspired racetrack where kids turn interchangeable gears to drive a toy car. I built the electronics (a speaker with an I2S amplifier, a break beam sensor, and an addressable LED strip, all controlled by a Raspberry Pi Pico), wrote the code, designed the parts in AutoCAD, 3D-printed and laser-cut the components, wrote the final report, and assembled and painted the exhibit. Programmed in MicroPython, the Pico polls the break beam sensor every 50 ms; when the car passes, it streams a WAV file I created of "Life is a Highway" to a MAX98357A class D amplifier while animating the LED strip, updating the LEDs every 8th audio chunk so the Pico isn't overloaded. In AutoCAD, I designed the enclosures, internal supports, and nine interchangeable plywood spur gears (4, 9, and 14 cm), which children mount on pegs to drive a string-and-pulley loop that moves the car.

★ Museum test, April 12, 2026: 102 s avg. engagement (goal: 90 s) · 6 gears used on avg. (goal: 3) · $111.03 of $120 budget

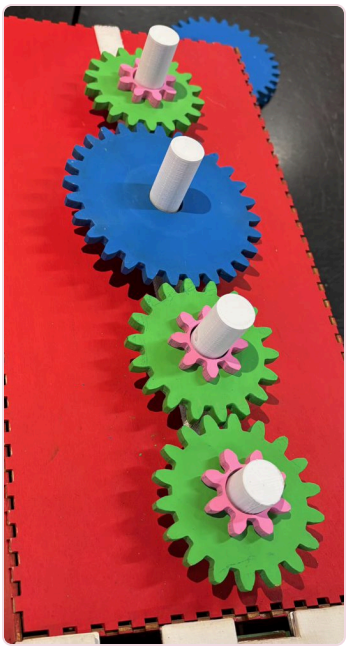
Tools: MicroPython · Raspberry Pi Pico · I2S Audio · LED Strip · Break Beam Sensor · AutoCAD · 3D Printing · Laser Cutting



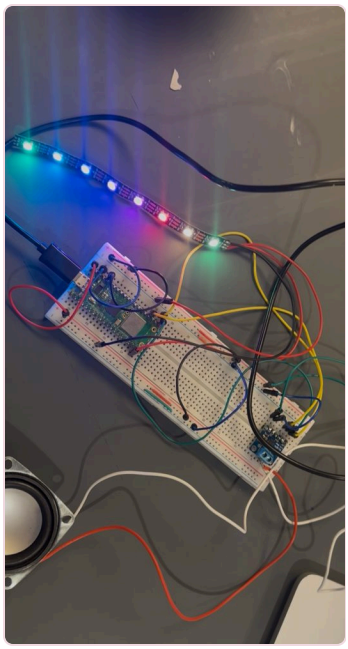
Exhibit at the museum



Gears and toy cars



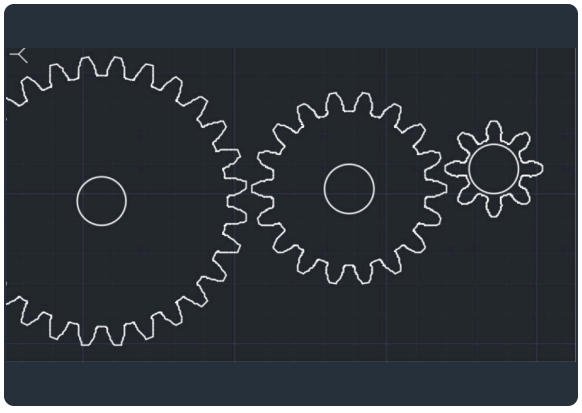
Interchangeable gears



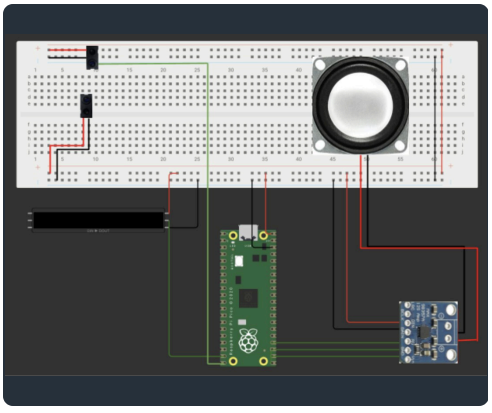
Electronics prototype



SolidWorks pulley design



AutoCAD gear design



Pico, LED strip, sensor, and speaker wiring

Intelligent Urban Infrastructure

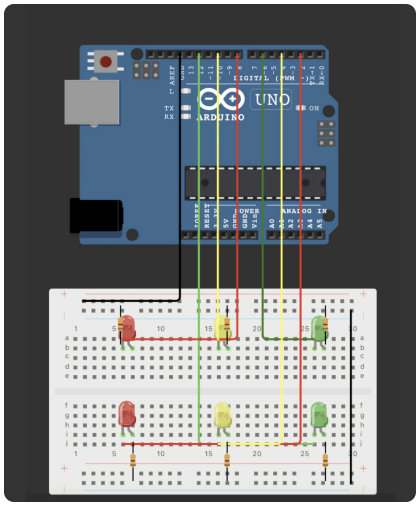
2022 – Present

Independent Research · IoT Traffic-Light Model

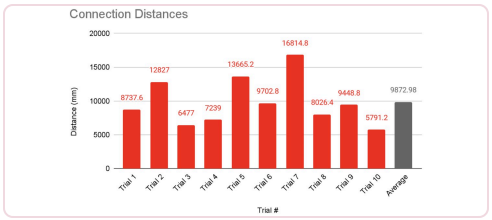
Since 2022, I have been designing and building an IoT traffic-light model to reduce the energy wasted by traffic lights that are always on. In the first year, I built a two-light intersection controlled by an Arduino Uno WiFi Rev2 that turns on only when a smartphone, representing a car, connects over Bluetooth Low Energy. Across 10 trials, I recorded the connection and disconnection distances, and the lights turned on and off with every successful connection. In the second year, I added a pedestrian component: a second Arduino with a GPS module sends its speed to the traffic light over WiFi using MQTT, and the light gives the green first to the pedestrian or to the vehicle depending on the speed it receives. The model worked without error in all 20 trials (10 vehicle and 10 pedestrian). I am continuing to expand this research. I recently built an interactive simulation of a demand-responsive intersection that stays dark until a connected car or pedestrian is in range, using my prototype's measured Bluetooth range and GPS speed data to compare its energy use against an always-on signal.

★ 3rd Place (2023) and 2nd Place (2024), MA Regional Science Fair · US Air Force Award · Advanced to state competitions

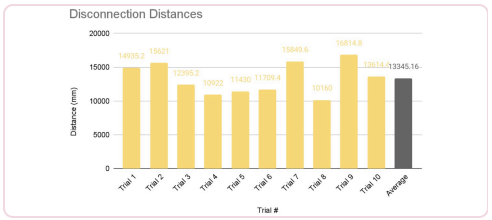
Tools: Arduino · C++ · Bluetooth Low Energy · WiFi · MQTT · GPS | **Live simulation:** ludomelodia.github.io/smart-intersection



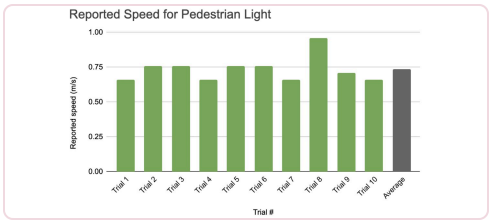
Year 1 circuit design



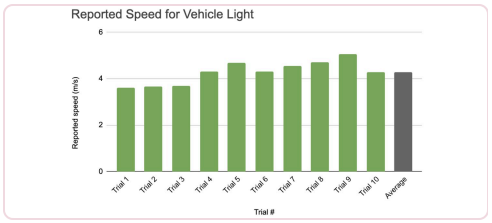
Year 1: Bluetooth connection distance per trial



Year 1: Bluetooth disconnection distance per trial



Year 2: GPS-reported speed, pedestrian trials



Year 2: GPS-reported speed, vehicle trials

The Train Game

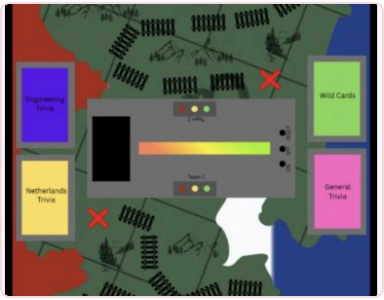
October – December 2025

GE 1501 Cornerstone of Engineering, Northeastern University · Three-person team · Client project

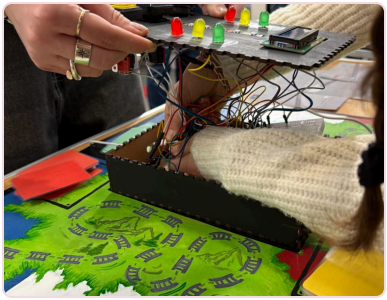
I built all of the electronics hardware for a custom strategy game, wiring three push buttons, six LEDs arranged as two traffic lights, an addressable LED strip, and a 16×2 I2C LCD to a Raspberry Pi Pico housed in a laser-cut wooden box, and I helped write the code. The system runs 45-second timed rounds, displaying the countdown on the LCD and traffic lights, while the LED strip randomly selects which of four trivia decks each team answers from. Our team built it for a client, a civil engineering student interested in transportation and the Netherlands, within a $100 budget and an eight-week timeline. We translated his needs into functions and constraints (2–6 players, 10–15 minute games, portability, and durability) and ranked design traits using a pairwise comparison chart.

★ Chosen by the client as his favorite project

Tools: Raspberry Pi Pico · MicroPython · Circuit Wiring · I2C LCD · Addressable LED Strip · LEDs · Push Buttons · Laser Cutting



Digital design render



Electronics and wiring



Final board with controls



Presentation demo

Autonomous Underwater Vehicle Challenge

July – August 2024

MIT Beaver Works Summer Institute · Instructor: Mohamed Saad Ibn Seddik

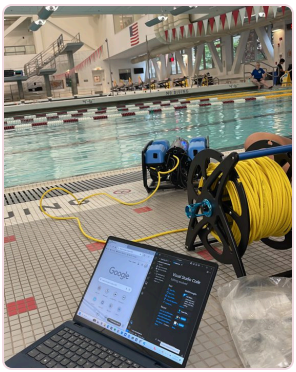
In MIT Beaver Works Summer Institute's four-week Autonomous Underwater Vehicle Challenge, my team programmed a BlueROV2 to navigate a pool without human control. Using Python and ROS, we built separate nodes for depth control, lane following, and AprilTag detection, each feeding PID controllers that drove the vehicle's thrusters. To help the AUV tell which side of the pool it was on, we placed AprilTag markers (similar to QR codes) on every side of the pool, anchoring them underwater by attaching them to rocks we found outside MIT, which earned our team the Master Navigator Award. In the final challenge, our AUV went head-to-head against another team's: both vehicles searched the pool, and the first to detect the other with its camera flashed its lights. Our AUV found the other first.

★ Master Navigator Award

Tools: Python · ROS · PID Control · Computer Vision · AprilTags · BlueROV2



AprilTags anchored with rocks



Pool testing



Inside the BlueROV2



Our team with the AUV

GEO Engineers

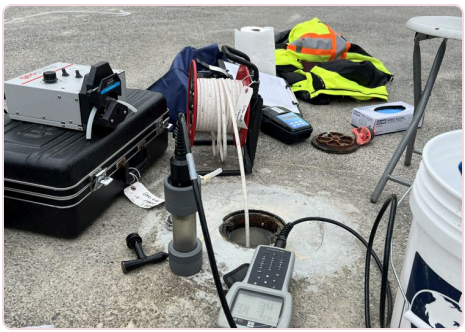
April – June 2025

Engineering Intern · Boston, MA

I contributed to geotechnical engineering projects at GEO Engineers, a Boston-based firm, across both office and field work. My work included soil and material sampling, site investigations, boring logs, and geotechnical drawings.



Field work



Sampling equipment



On site

Forge: Cooking Robot

September 2026 – Present

Hardware Product Lab Member · Team Electrical Engineer · A Sherman Center Program, Northeastern University

As the team's electrical engineer, I program the Arduino, assemble the electronics, and work on PCB design in KiCad to customize the board for a cooking robot that stores dry ingredients and, at the press of a button, dispenses them in preset measurements. The product idea was mine. Forge is Northeastern's student-run product development studio, where members work on a semester-long team project spanning electronics, CAD, and coding. *In progress.*

Tools: Arduino · KiCad · PCB Design · Electronics Assembly

Full project reports and research papers: ludomelodia.github.io